

116 Advanced Statistics for Neuroscience

Exercise 05

Benjamini-Hochberg

FDR

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Correct for false discovery rate (FDR)

When you have many data sets to analyze, some statistical tests may turn out significant (reject the null-hypothesis) purely by chance.

To limit the probability of **false discovery**, you must apply some stricter statistical standard. Here you will apply the B-H method for FDR correction, which is the accepted standard.

Specifically, you will

- ▶ Obtain the values of some statistics from a supplied function.
- ▶ Obtain the true nature of each of these statistics (null-hypothesis or not) from a supplied oracle.
- ▶ Plot the cumulative distribution of these values.
- ▶ Plot the inverse cumulative distribution and compare both the B-H threshold and the true threshold (supplied by the oracle).

Obtain samples

The supplied Python-function will generate samples from a Gaussian mixture. Whereas most samples derive from a standard normal distribution (i.e., conform to the null-hypothesis), a minority of samples derives from a normal distribution with larger mean (i.e., do not conform to the null-hypothesis).

In addition, this function will serve as an oracle and reveal the source of every individual sample (null or non-null), as well as the critical threshold value, above which samples are more likely to be non-null, and below which samples are more likely to be null.

$$(\text{samples}, \text{sources}, \text{sthresh}) = \text{MakeSamples}(\text{Nsample})$$

where Nsample is the number of samples, samples is an array of sample values, $\text{sources} \in \{0, 1\}$ is an array of the source identities, and sthresh is the critical threshold value.

Cumulative distribution function

Sort your sample values, starting with the **largest value**, and arrange the source identities in the same order.

Generate a cumulative probability vector, $(1 : N_{\text{sample}})/N_{\text{sample}}$.

Plot the sample values along the x-axis and the cumulative probability values along the y-axis. The largest samples should have the smallest probability.

Using the source identifiers, replot the non-null hypothesis samples in red.

Discussion:

Please explain why these red samples are concentrated in a particular region of the distribution?

Benjamini-Hochberg plot

Convert all sample values to p-values by computing the **tail area** of a standard normal distribution, i.e., the area between the sample value and positive infinity.

Plot cumulative probability along the x-axis and p-values along the y-axis. Limit your plotting range to $[0, 0.05]$ along both axes. The plotted symbols should start at the lower left (near the origin) and gradually increase towards the upper right.

As before, plot the symbols for **non-null samples** in red!

Add to your plot a horizontal and **black dashed** line at the p-value of your true criterion threshold.

Benjamini-Hochberg criterion

True criterion

Add to your plot a horizontal line with the p-value of your true criterion threshold.

B-H criterion

Finally, add to your plot the B-H-criterion: a straight and **magenta-colored** line with a slope of $q = +0.2$.

q is the fraction of **false** discoveries that you are willing to accept.

Discussion

Please explain why many, but not all, red symbols are below both criteria?

Double check

Increase the number of samples and check your fraction of false discoveries. How does it compare to q ?